BUDGET JUSTIFICATION
The Pennsylvania State University

(A-B.) Personnel ($27,797) – The principal investigator is budgeted at the percentage of time shown using his/her actual salary in the calculation. The principal investigator's time includes both technical and project management functions. Any other individuals/positions shown are technical staff with the percentage of time shown and actual salaries used. For project time occurring after July 1 of any given year, the salaries have been adjusted at the University approved rate of 4%.

Senior/Key Personnel ($3,047):

- **Elia Merzari, Principal Investigator** – 9.9% effort over 1.5 months (.15 summer months) for the project. Elia Merzari will support the project as the PI at Penn State, interacting with project leadership and supervising the activities of the graduate student. He will also help identify and define benchmark problems relevant to nuclear energy and support the broader goals of the project.

Other Personnel ($24,750):

- **To Be Named, Graduate Assistant** – One Grade 16 Graduate Assistant @ 50% effort over 7.5 months (3.75 academic months) for the project. The student will perform the majority of the work at Penn State, focusing on the development of high-fidelity numerical models relevant to nuclear energy and on testing the novel foundation-model approaches developed in the project. The student will also run simulations, post-process results, and prepare documents, presentations, and reports as required by project leadership.
  ** Graduate Assistants at Penn State University are considered full-time at 50% effort with the expectation that the remaining 50% is dedicated to coursework.*

Fringe Benefits ($4,009) – Fringe benefits are computed using the fixed rates of 40.6% applicable to Category I Salaries, 11.2% applicable to Category II Graduate Assistants, 7.9% applicable to Category III Wages, 0.3% applicable to Category IV Student Wages, and 27.7% for Category V, Postdoctoral Scholars and Fellows, for fiscal year 2026 (July 1, 2025, through June 30, 2026). If this proposal is funded, the rates quoted above shall, at the time of funding, be subject to adjustment for any period subsequent to June 30, 2026, if superseding Government approved rates have been established. Fringe benefit rates are negotiated and approved by the Office of Naval Research, Penn State's cognizant federal agency.

(C.) Equipment ($0)

(D.) Travel ($10,000) – All travel will be in accordance with University travel regulations and mileage will be charged at the current rate on the date of travel. Travel estimates are based on costs that were incurred on previous projects of a similar nature for federal and state agencies.

1. **Domestic Travel ($10,000)** – Funding is requested for the PI to attend the required two annual Genesis Mission PI Meetings. The PI will participate in coordination activities with other projects and to effectively communicate with other researchers. Location TBD.

Year	Destination	Dates	# Travelers	Cost per Traveler	Total Estimated Costs	Purpose/Relation to Project	Basis for Cost Estimate
2026	TBD	TBD	1	$5,000	$5,000	annual PI meeting	Prior travel
2027	TBD	TBD	1	$5,000	$5,000	annual PI meeting	Prior travel

(E.) Participant/Trainee Support Costs ($0)

(F.) Other Direct Costs

1. **Materials and Supplies ($0)**
2. **Publication Costs ($0)**
3. **Consultant Services ($0)**
4. **ADP/Computer Services ($0)**
5. **Subawards/Consortium/Contractual Costs ($0)**
6. **Equipment or Facility Rental/User Fees ($0)**
7. **Alterations and Renovations ($0)**
8. **Other ($4,242)**:
    - **Graduate Assistant Tuition ($4,242)** – Computed using the approved tuition charges for a one-half (1/2) time graduate assistant of $10,810 for fall semester 2025, $10,810 for spring semester 2026, and $5,515 for summer session 2026. The charges quoted above are increased by two percent (2%) for any project period occurring after summer session 2026, and each summer session thereafter. NOTE*:* It is anticipated that the graduate student assigned to this project will be classified as a post-comprehensive [level 601] PhD student, therefore a reduced dissertation fee has been budgeted for this project.
      ** Tuition is included for Graduate Assistants for Fall and Spring Semesters.*

(G.) Total Direct Costs (A-F)
Year 1: $46,048; Total: $46,048

(H.) Indirect Costs (Facilities and Administrative Costs) ($23,952) – F&A rates are negotiated and approved by the Office of Naval Research, Penn State's cognizant federal agency. Penn State's current fixed on-campus rate for research is 57.3% of MTDC from July 1, 2025, through June 30, 2026. New awards and new competitive segments with an effective date of July 1, 2026, or later shall be subject to adjustment when superseding Government approved rates are established. Per 2 CFR 200 (Appendix III, Section C.7), the actual F&A rates used will be fixed at the time of the initial award for the duration of the competitive segment.

Cognizant Agency: Office of Naval Research
Contact Information: Boston Regional Office
495 Summer Street
Boston, MA 02210
Email: usn.pentagon.cnr-arlington-va.mbx.onr-boston@us.navy.mil

Year 1: $23,952; Total: $23,952

(I.) Total Direct and Indirect Costs (G-H)
Year 1: $70,000; Total: $70,000

The Pennsylvania State University employs individuals and accepts students and graduate research students from a multitude of national backgrounds. As an entity, the University is subject to, and works diligently to obey, federal regulations regarding the export of controlled technologies and data. Sponsor, as an independent entity, is individually responsible for ascertaining its compliance with federal export laws and procedures. If Sponsor anticipates disclosure or provision of controlled technology or data to University as part of the proposed sponsored project, Sponsor should inform University, in writing, of the existence of, and information concerning the scope and extent of, such anticipated disclosures or provisions.

DEPARTMENT OF THE NAVY
OFFICE OF NAVAL RESEARCH
875 NORTH RANDOLPH STREET
SUITE 1425
ARLINGTON, VA 22203-1995

IN REPLY REFER TO:

February 26, 2026
[Supersedes Rate Agreement dated June 10, 2025]

NEGOTIATION AGREEMENT

INSTITUTION: **THE PENNSYLVANIA STATE UNIVERSITY**
UNIVERSITY PARK, PA 16801-3857

The Facilities and Administrative (F&A) cost rates contained herein are for use on grants, contracts and/or other agreements issued or awarded to the Pennsylvania State University (PSU) by all Federal Agencies of the United States of America, in accordance with the provisions and cost principles mandated by 2 CFR Part 200. These rates shall be used for forward pricing and billing purposes for the PSU's Fiscal Year 2026. This rate agreement supersedes all previous rate agreements/determinations related to these rates for Fiscal Year 2026.

SECTION I – RATES – TYPE: FIXED (FIXED)

UNIVERSITY PARK (F&A)

TYPE	FROM	TO	RATE	BASE	APPLICABLE TO	LOCATION
FIXED	07/01/25	06/30/26	57.3%	(a)	Organized Research (1,4)	On-Campus
FIXED	07/01/25	06/30/26	26.0%	(a)	Organized Research (1,4)	Off-Campus
FIXED	07/01/25	06/30/26	64.4%	(a)	Organized Research (2,4)	On-Campus
FIXED	07/01/25	06/30/26	33.1%	(a)	Organized Research (2,4)	Off-Campus
FIXED	07/01/25	06/30/26	53.3%	(a)	Instruction (1,4)	On-Campus
FIXED	07/01/25	06/30/26	26.0%	(a)	Instruction (1,4)	Off-Campus
FIXED	07/01/25	06/30/26	24.1%	(a)	OSA[1] (1,4)	On-Campus
FIXED	07/01/25	06/30/26	16.8%	(a)	OSA (1,4)	Off-Campus
FIXED	07/01/25	06/30/26	23.7%	(a)	OSA HHS (1,3)	On-Campus
FIXED	07/01/25	06/30/26	16.4%	(a)	OSA HHS (1,3)	Off-Campus

APPLIED RESEARCH LABORATORY

TYPE	FROM	TO	RATE	BASE	APPLICABLE TO	LOCATION
FIXED	07/01/25	06/30/26	6.5%	(b)	Organized Research – ARL[2]	On-Campus
FIXED	07/01/25	06/30/26	5.3%	(b)	Organized Research - ARL	Off-Campus

HERSHEY COLLEGE OF MEDICINE

TYPE	FROM	TO	RATE	BASE	APPLICABALE TO	LOCATION
FIXED	07/01/25	06/30/26	56.6%	(a)	Organized Research – HCM[3]	On-Campus
FIXED	07/01/25	06/30/26	26.0%	(a)	Organized Research - HCM	Off-Campus

[1]Other Sponsored Activities
[2]Applied Research Laboratory
[3] Hershey College of Medicine

DISTRIBUTION BASE

(a) Modified Total Direct Cost (MTDC) means all direct salaries and wages, applicable fringe benefits, materials and supplies, services, travel, and up to the first $50,000 of each subaward (regardless of the period of performance of the subawards under the award). MTDC excludes equipment, capital expenditures, charges for patient care, rental costs, tuition remission, scholarships and fellowships, participant support costs, and the portion of each subaward in excess of $50,000.

(b) The ARL sponsored base consists of all on-site and off-site direct costs recorded to projects, (excluding direct material and equipment related costs and direct subcontract costs and tuition). The ARL internal overhead base consists of all on-site and off-site ARL internal overhead costs and F&A is allocated on all direct costs (excludes depreciation, rent, tuition, and allowable interest).

APPLICABLE TO

(1) Applies to DOD contracts awarded before November 30, 1993, all Non-DOD Instruments, and all DOD grants and other agreements (See Section II, paragraph E). (Capped).

(2) Applies to only DOD contracts awarded on or after November 30, 1993 in accordance with and under the authority of DFARS 231.303(1) (See Section II, paragraph E). (Uncapped).

(3) Applies to all awards subject to Executive Level II Salary Limitations as defined in the provisions of HHS Appropriation Act (PUB.L. No. 117-328, div. H, title II) or similar provision in subsequent appropriations acts.

(4) Applies to all awards not subject to the HHS Appropriation.

SECTION II - GENERAL TERMS AND CONDITIONS

A. **LIMITATIONS**: Use of the rates set forth under Section I is subject to availability of funds and to any other statutory or administrative limitations. The rates are applicable to a given grant, contract or other agreement only to the extent that funds are available and consistent with any and all limitations of cost clauses or provisions, if any, contained therein. Acceptance of any or all of the rates agreed to herein is predicated upon the following conditions: (1) that no

costs other than those incurred by the institution were included in this indirect cost pool as finally accepted and that such costs are legal obligations of the institution and allowable under governing cost principles; (2) that the same costs that have been treated as indirect costs are not claimed as direct costs; (3) that similar types of costs have been accorded consistent accounting treatment; and (4) that the information provided by the institution which was used as a basis for acceptance of the rates agreed to herein, and expressly relied upon by the Government in negotiating and accepting the said rates is not subsequently found to be materially incomplete or inaccurate.

B. **ACCOUNTING CHANGES**: The rates contained in Section I of this agreement are based on the accounting system in effect at the time the agreement was negotiated. Changes to the method(s) of accounting for costs, which affect the amount of reimbursement resulting from the use of these rates require the prior written approval of the authorized representative of the cognizant agency for indirect costs. Such changes include but are not limited to changes in the charging of a particular type of cost from indirect to direct. Failure to obtain such approval may result in subsequent cost disallowances.

C. **FIXED RATES WITH CARRY-FORWARD PROVISIONS**: The fixed rates contained in this agreement are based on estimates of the costs for FY 2026. When actual costs for this fiscal year are determined, adjustments will be applied to a rate negotiation for a subsequent fiscal year to recognize the difference between the FY 2026 estimated costs used to establish the fixed rates and the negotiated actual FY 2026 costs.

D. **CARRY-FORWARD AMOUNTS**: The following estimated CFW for FY24 carry-forward amounts were included in the establishment of the FY2026 rates: ( ) Denotes over-recovery.

FY24 Estimated CFW

Organized Research – University Park	Liquidated in FY 26
Organized Research On-Campus Capped	($3,896,035)
Instruction On Campus	$17,773,801
OSA On Campus	($1,214,696)
OSA Off Campus	($28,851)
ARL On Campus	$205,096
ARL Off Campus	($15,152)
Hershey	($2,812,595)

E. **USE BY OTHER FEDERAL AGENCIES**: The rates set forth in Section I are negotiated in accordance with and under the authority set forth in 2 CFR Part 200. Accordingly, such rates shall be applied to the extent provided in such regulations to grants, contracts, and other agreements to which 2 CFR Part 200 applies, subject to any limitations in part A of this section. Copies of this document may be provided by either party to other federal agencies to provide such agencies with documentary notice of this agreement and its terms and conditions.

F. **DFARS WAIVER**: Signature of this agreement by the authorized representative of Pennsylvania State University and the Government acknowledges and affirms the University's request to waive the prohibition contained in DFARS 231.303(1) and the Government's exercise of its discretion contained in DFARS 231.303(2) to waive the prohibition in DFARS 231.303(1) with regards to all the rates except F&A Organized Research. The waiver request by Pennsylvania State University is made to simplify the University's overall management of DoD cost reimbursements under DoD contracts.

G. **SPECIAL REMARKS**:

1. The rates included in Section I are not intended to be applied to Intergovernmental Personnel Act (IPA) costs. If the Pennsylvania State University elects to seek reimbursement of F&A or internal overhead costs associated with IPA agreements, then the University and the Office of Naval Research shall establish special F&A and/or internal overhead rates for IPA agreements in accordance with the provisions of 2 CFR Part 200.
2. The Government's agreement with the rates set forth in Section I is not an acceptance of Pennsylvania State University's accounting practices or methodologies. Any reliance by the Government on cost data or methodologies submitted by Pennsylvania State University is on a non-precedence-setting basis and does not imply Government acceptance.

Accepted:

FOR PENNSYLVANIA STATE UNIVERSITY:	FOR THE U.S. GOVERNMENT:
Sara Thorndike Senior Vice President for Finance & Business/Treasurer	Betty J. Tingle Contracting Officer
2/26/2026 Date	Date

For information concerning this agreement contact:

Betty Tingle, Contracting Officer
Office of Naval Research

Phone: (948) 215-0521
E-mail: betty.j.tingle.civ@us.navy.mil

DEPARTMENT OF THE NAVY
OFFICE OF NAVAL RESEARCH
875 NORTH RANDOLPH STREET
SUITE 1425
ARLINGTON, VA 22203-1995

IN REPLY REFER TO:

Agreement Date: June 27, 2025
[Supersedes Agreement Dated: June 27, 2024]

NEGOTIATION AGREEMENT

INSTITUTION: **THE PENNSYLVANIA STATE UNIVERSITY**
UNIVERSITY PARK, PA 16801-3857

The Fringe Benefit rates contained herein are for use on grants, contracts and/or other agreements issued or awarded to the Pennsylvania State University (PSU) by all Federal agencies of the United States of America in accordance with the provisions and cost principles mandated by 2 CFR Part 200. These rates shall be used for forward pricing and billing purposes for the PSU's Fiscal Years 2026 and 2027. This rate agreement supersedes all previous rate agreements/determinations related to these rates for Fiscal Year 2026.

Section I: RATES - TYPE: FIXED (Fixed) and PROVISIONAL (Prov)

TYPE	FROM	TO	RATE	BASE	APPLICABLE TO	LOCATION
Fixed	7/1/2025	6/30/2026	40.6%	(a)	Salaries	All
Fixed	7/1/2025	6/30/2026	7.9%	(b)	Wages	All
Fixed	7/1/2025	6/30/2026	11.2%	(c)	Graduate Assistants	All
Fixed	7/1/2025	6/30/2026	0.3%	(d)	Students	All
Fixed	7/1/2025	6/30/2026	27.7%	(e)	Postdoctoral Students	All
Prov	7/1/2026	6/30/2027	40.6%	(a)	Salaries	All
Prov	7/1/2026	6/30/2027	7.9%	(b)	Wages	All
Prov	7/1/2026	6/30/2027	11.2%	(c)	Graduate Assistants	All
Prov	7/1/2026	6/30/2027	0.3%	(d)	Students	All
Prov	7/1/2026	6/30/2027	27.7%	(e)	Postdoctoral Students	All

DISTRIBUTION BASES

(a) Salaries, excluding graduate assistants, postdoctoral scholars, fixed term II and visiting faculty appointments not receiving full benefits.
(b) Wages and those salaries not receiving full benefits, exclusive of graduate assistants, student wages, and postdoctoral scholar salaries.
(c) Graduate assistant salaries.
(d) Student wages.
(e) Postdoctoral Scholar salaries.

SECTION II - GENERAL TERMS AND CONDITIONS

A. **LIMITATIONS:** Use of the rates set forth under Section I is subject to availability of funds and to any other statutory or administrative limitations (e.g., award term with indirect rate cap). The rates are applicable to a given grant, contract or other agreement only to the extent that funds are available and consistent with any and all limitations of cost clauses or provisions, if any, contained therein. Acceptance of any or all of the rates agreed to herein is predicated upon the following conditions: (1) that no costs other than those incurred by the institution were included in this indirect cost pool as finally accepted and that such costs are legal obligations of the institution and allowable under governing cost principles; (2) that the same costs that have been treated as indirect costs are not claimed as direct costs; (3) that similar types of costs have been accorded consistent accounting treatment; and (4) that the information provided by the institution which was used as a basis for acceptance of the rates agreed to herein, and expressly relied upon by the Government in negotiating and accepting the said rates is not subsequently found to be materially incomplete or inaccurate.

B. **ACCOUNTING CHANGES:** The rates contained in Section I of this agreement are based on the accounting system in effect at the time the agreement was negotiated. Changes to the method(s) of accounting for costs, which affect the amount of reimbursement resulting from the use of these rates require the prior written approval of the authorized representative of the cognizant agency for indirect costs. Such changes include but are not limited to changes in the charging of a particular type of cost from indirect to direct. Failure to obtain such approval may result in subsequent cost disallowances.

C. **FIXED RATES WITH CARRY-FORWARD PROVISIONS:** The fixed rates in this agreement are based on estimates of the costs for FY 2026. When actual costs for this fiscal year are determined, adjustments will be applied to a rate negotiation for a subsequent fiscal year to recognize the difference between the FY 2026 estimated costs used to establish the fixed rates and the negotiated actual FY 2026 costs.

D. **CARRY FORWARD AMOUNTS:** The estimated carry-forwards for FY24 were included in the establishment of the FY 2026 rates:

Liquidated in FY26

Employee Category	FY24 Estimated CFW Liquidated in FY26 Fixed Rates
Salaries	$23,330,634
Wages	$1,695
Graduate Assistants	$192,516
Student Wages	8,377
Postdoctoral Scholar Salaries	$50,565

E. **PROVISIONAL RATES:** The provisional rates contained in this agreement are subject to unilateral amendment by the Government or bilateral amendment by the contracting parties at any time.

F. **USE BY OTHER FEDERAL AGENCIES**: The rates set forth in Section I are negotiated in accordance with and under the authority set forth in 2 CFR Part 200. Accordingly, such rates shall be applied to the extent provided in such regulations to grants, contracts, and other agreements to which 2 CFR Part 200 applies, subject to any limitations in part A of this section. Copies of this document may be provided by either party to other federal agencies to provide such agencies with documentary notice of this agreement and its terms and conditions.

Accepted:

FOR PENNSYLVANIA STATE UNIVERSITY:	FOR THE U.S. GOVERNMENT:
Sara Thorndike, Senior Vice President for Finance & Business/Treasurer	Betty J. Tingle Contracting Officer
7/1/25 Date	Date

For information concerning this agreement contact:
Betty Tingle, Contracting Officer, Phone: (703) 696-7742, E-mail: betty.j.tingle.civ@us.navy.mil